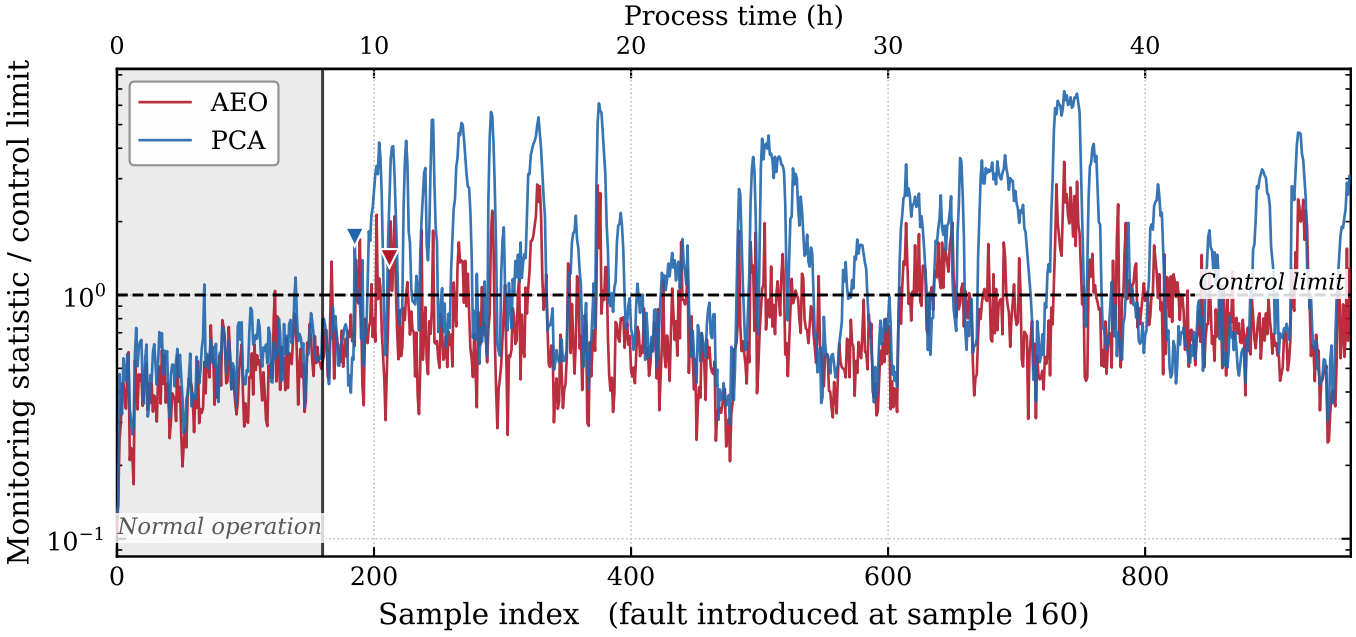
Fault Detection Performance for IDV(16)

Combined T^2 and SPE statistics — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	80.0 samples (240 min)	MDR 0.761	detected in 7/7 runs
PCA	delay	21.0 samples (63 min)	MDR 0.428	detected in 7/7 runs

PCA detects 59.0 samples (177 min) earlier; PCA missed-detection rate lower by 44%